

## WEEK 1 TASK: DATA CLEANING AND PRELIMINARY ANALYSIS REPORT

### 1. Project Overview

- \* Intern Name: Ananya Rastogi
- \* Role: Virtual R Data Analyst Intern
- \* Organization: YuvaIntern
- \* Dataset Used: Olist Customers Dataset (olist\_customers\_dataset.csv)

### 2. Introduction

For the Week 1 task, the Olist Customers Dataset was selected to perform basic data ingestion, structure inspection, and preliminary analysis using R Programming. The primary objective was to check the structure, generate summary statistics, and identify any missing values within the workspace environment.

### 3. R Code Implemented

The following clean R script was executed in the Posit Cloud console to perform the analysis:

```
data <- read.csv("olist_customers_dataset.csv")
```

```
print("--- DATA STRUCTURE ---")
```

```
str(data)
```

```
print("--- DATA SUMMARY ---")
```

```
summary(data)
```

```
print("--- MISSING VALUES COUNT ---")
```

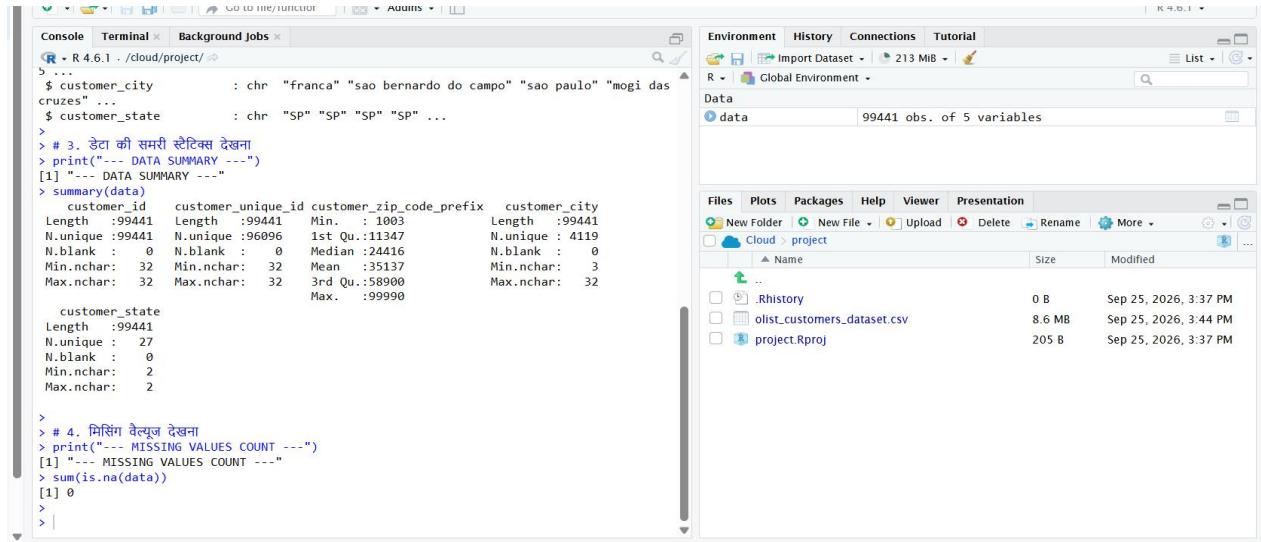
```
sum(is.na(data))
```

### 4. Key Observations & Summary Statistics

- \* Data Dimensions: The dataset contains a total of 99,441 records tracking customer locations across Brazil.
- \* Geographical Distribution: Customers are mapped across 27 unique states (customer\_state) and 4,119 unique cities (customer\_city).

\* Data Integrity: The execution of the `sum(is.na(data))` function returned a result of 0, confirming that there are no missing values (NA values) present in the dataset.

## 5. Output Evidence



The screenshot displays the RStudio environment with the following components:

- Console:** Shows the execution of R code. The first part displays the structure of the `data` object, including variables like `customer_city`, `customer_state`, `customer_id`, `customer_unique_id`, `customer_zip_code_prefix`, and `customer_city`. The second part shows the output of `summary(data)`, providing statistical summaries for each variable. The third part shows the output of `sum(is.na(data))`, which returns 0, indicating no missing values.
- Environment:** Shows the `data` object with 99441 observations and 5 variables.
- Files:** Lists the files in the project directory, including `.Rhistory`, `olist_customers_dataset.csv`, and `project.Rproj`.

```
> ...  
$ customer_city      : chr "franca" "sao bernardo do campo" "sao paulo" "mogi das  
cruzes" ...  
$ customer_state     : chr "sp" "sp" "sp" "sp" ...  
>  
> # 3. डेटा की समरी स्टैटिक्स देखना  
> print("--- DATA SUMMARY ---")  
[1] "--- DATA SUMMARY ---"  
> summary(data)  
  customer_id customer_unique_id customer_zip_code_prefix customer_city  
Length :99441 Length :99441 Min. : 1003 Length :99441  
N.unique :99441 N.unique :96096 1st Qu.:11347 N.unique : 4119  
N.blank : 0 N.blank : 0 Median :24416 N.blank : 0  
Min.nchar: 32 Min.nchar: 32 Mean :35137 Min.nchar: 3  
Max.nchar: 32 Max.nchar: 32 3rd Qu.:58900 Max.nchar: 32  
Max. :99990  
  
  customer_state  
Length :99441  
N.unique : 27  
N.blank : 0  
Min.nchar: 2  
Max.nchar: 2  
  
>  
> # 4. मिसिंग वैल्यूज देखना  
> print("--- MISSING VALUES COUNT ---")  
[1] "--- MISSING VALUES COUNT ---"  
> sum(is.na(data))  
[1] 0  
>  
>
```

## 6. Conclusion

The data was successfully processed in the R environment. Preliminary operations such as structural validation and summary analysis confirm that the dataset is clean, structurally sound, and fully prepared for advanced predictive modeling or visualization in the upcoming weeks.